White Papers & Architectures

Well Architected Framework, Disaster Recovery, etc…

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Section Overview

- Well Architected Framework Whitepaper
- Well Architected Tool
- AWS Trusted Advisor
- Reference architectures resources (for real-world)
- Disaster Recovery on AWS Whitepaper

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Well Architected Framework General Guiding Principles

- https://aws.amazon.com/architecture/well-architected
- Stop guessing your capacity needs
- Test systems at production scale
- Automate to make architectural experimentation easier
- Allow for evolutionary architectures
  - Design based on changing requirements
- Drive architectures using data
- Improve through game days
  - Simulate applications for flash sale days

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Well Architected Framework
6 Pillars

- 1) Operational Excellence
- 2) Security
- 3) Reliability
- 4) Performance Efficiency
- 5) Cost Optimization
- 6) Sustainability

- They are not something to balance, or trade-offs, they're a synergy

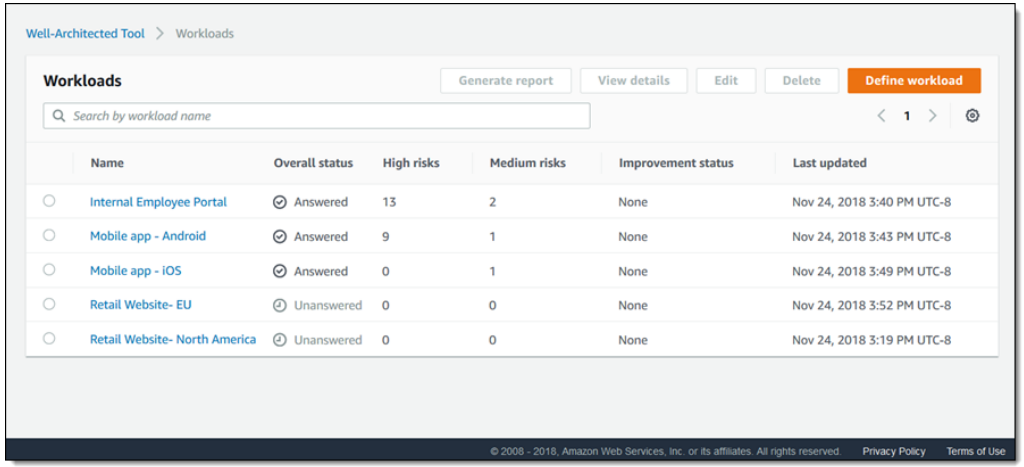
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Well-Architected Tool



- Free tool to **review your architectures** against the 6 pillars Well-Architected Framework and **adopt architectural best practices**
- How does it work?
  - Select your workload and answer questions
  - Review your answers against the 6 pillars
  - Obtain advice: get videos and documentations, generate a report, see the results in a dashboard
- Let's have a look: https://console.aws.amazon.com/wellarchitected



https://aws.amazon.com/blogs/aws/new-aws-well-architected-tool-review-workloads-against-best-practices/

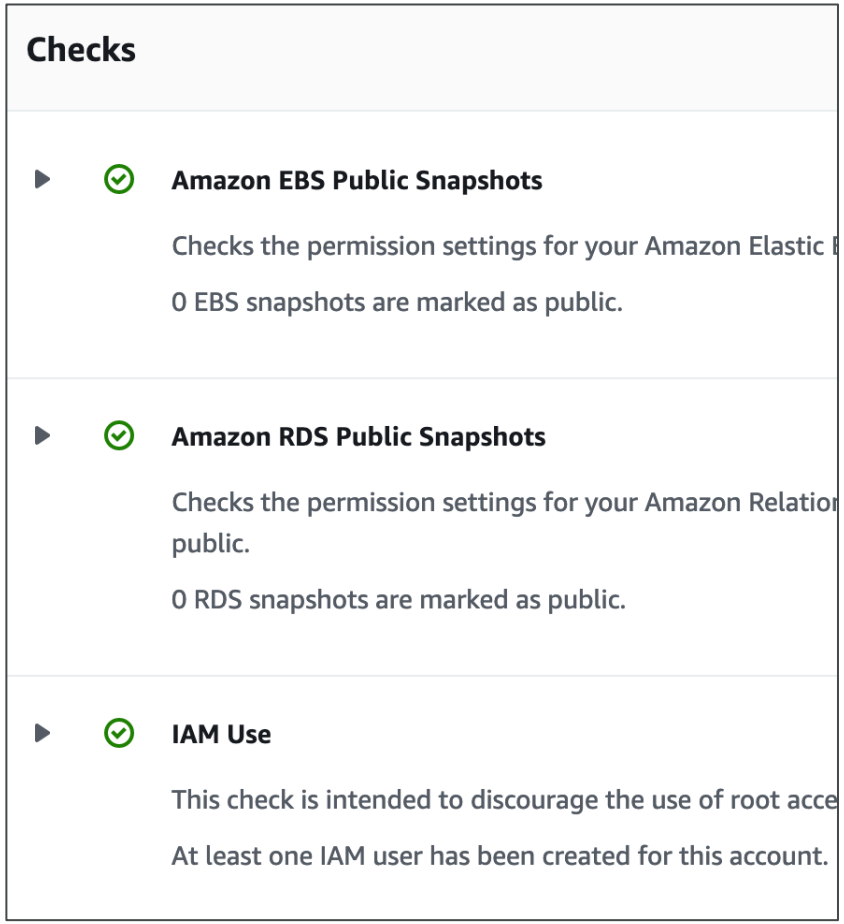
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Trusted Advisor



- No need to install anything – high level AWS account assessment
- Analyze your AWS accounts and provides recommendation on 6 categories:
  - Cost optimization
  - Performance
  - Security
  - Fault tolerance
  - Service limits
  - Operational Excellence
- Business & Enterprise Support plan
  - Full Set of Checks
  - Programmatic Access using AWS Support API



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

More Architecture Examples

- We've explored the most important architectural patterns:
  - **Classic:** EC2, ELB, RDS, ElastiCache, etc…
  - **Serverless:** S3, Lambda, DynamoDB, CloudFront, API Gateway, etc…

- If you want to see more AWS architectures:
- https://aws.amazon.com/architecture/
- https://aws.amazon.com/solutions/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Exam Review & Tips

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

State of learning checkpoint

- Let's look how far we've gone on our learning journey

- https://aws.amazon.com/certification/certified-solutions-architect-associate/

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Practice makes perfect

- If you're new to AWS, take a bit of AWS practice thanks to this course before rushing to the exam
- The exam recommends you to have one or more years of hands-on experience on AWS
- Practice makes perfect!

- If you feel overwhelmed by the amount of knowledge you just learned, just go through it one more time

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Proceed by elimination

- Most questions are going to be scenario based
- For all the questions, rule out answers that you know for sure are wrong
- For the remaining answers, understand which one makes the most sense

- There are very few trick questions
- Don't over-think it
- If a solution seems feasible but highly complicated, it's probably wrong

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Skim the AWS Whitepapers

- You can read about some AWS White Papers here:
  - Architecting for the Cloud: AWS Best Practices
  - AWS Well-Architected Framework
  - AWS Disaster Recovery (https://aws.amazon.com/disaster-recovery/)

- Overall we've explored all the most important concepts in the course
- It's never bad to have a look at the whitepapers you think are interesting!

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Read each service's FAQ

- FAQ = Frequently asked questions
- Example: https://aws.amazon.com/vpc/faqs/

- FAQ cover a lot of the questions asked at the exam
- They help confirm your understanding of a service

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Get into the AWS Community

- Help out and discuss with other people in the course Q&A
- Review questions asked by other people in the Q&A
- Do the practice test in this section

- Read forums online
- Read online blogs
- Attend local meetups and discuss with other AWS engineers
- Watch re-invent videos on Youtube (AWS Conference)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

How will the exam work?

- You'll have to register online at https://www.aws.training/
- Fee for the exam is 150 USD
- Provide one identity documents (ID, Passport, details are in emails sent to you…)
- No notes are allowed, no pen is allowed, no speaking
- 65 questions will be asked in 130 minutes
- Use the "Flag" feature to mark questions you want to re-visit
- At the end you can optionally review all the questions / answers

- To pass you need a score of a least 720 out of 1000
- You will know within 5 days if you passed / failed the exams (most of the time less)
- You will know the overall score a few days later (email notification)
- You will not know which answers were right / wrong
- If you fail, you can retake the exam again 14 days later

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Your AWS Certification journey

Foundational

Knowledge-based certification for
foundational understanding of AWS Cloud.
No prior experience needed.



Associate

Role-based certifications that showcase your knowledge
and skills on AWS and build your credibility as an AWS Cloud professional.
Prior cloud and/or strong on-premises IT experience recommended.



Professional

Role-based certifications that validate advanced skills
and knowledge required to design secure, optimized,
and modernized applications and to automate processes on AWS.
2 years of prior AWS Cloud experience recommended.



Specialty

Dive deeper and position yourself as a trusted advisor to your
stakeholders and/or customers in these strategic areas.
Refer to the exam guides on the exam pages for recommended experience.



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Architecture

Architecture

Solutions Architect

Design, develop, and manage cloud infrastructure and assets, work with DevOps to migrate applications to the cloud



optional for IT/ cloud professionals

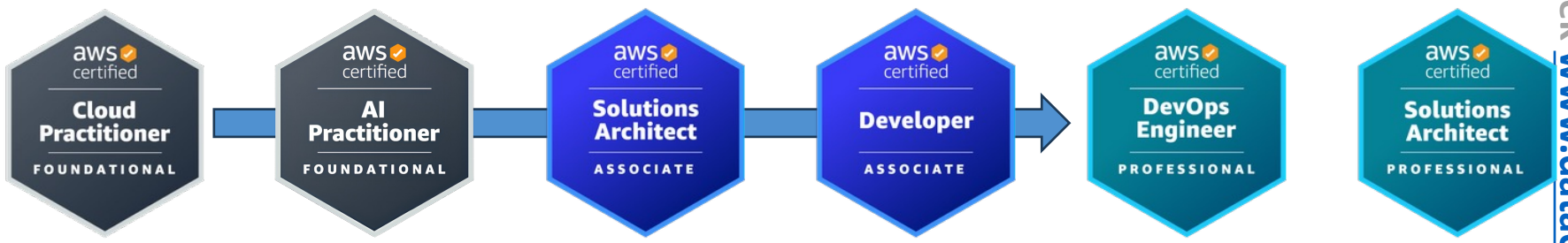
recommended for IT/cloud professionals to leverage AI

Dive Deep

Architecture

Application Architect

Design significant aspects of application architecture including user interface, middleware, and infrastructure, and ensure enterprise-wide scalable, reliable, and manageable systems



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Dive Deep

https://d1.awsstatic.com/training-and-certification/docs/AWS_certification_paths.pdf

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Operations

Operations

Systems Administrator

Install, upgrade, and maintain computer components and software, and integrate automation processes



optional for IT/ cloud professionals



Dive Deep

Operations

Cloud Engineer

Implement and operate an organization's networked computing infrastructure and Implement security systems to maintain data safety



optional for IT/ cloud professionals



Dive Deep

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

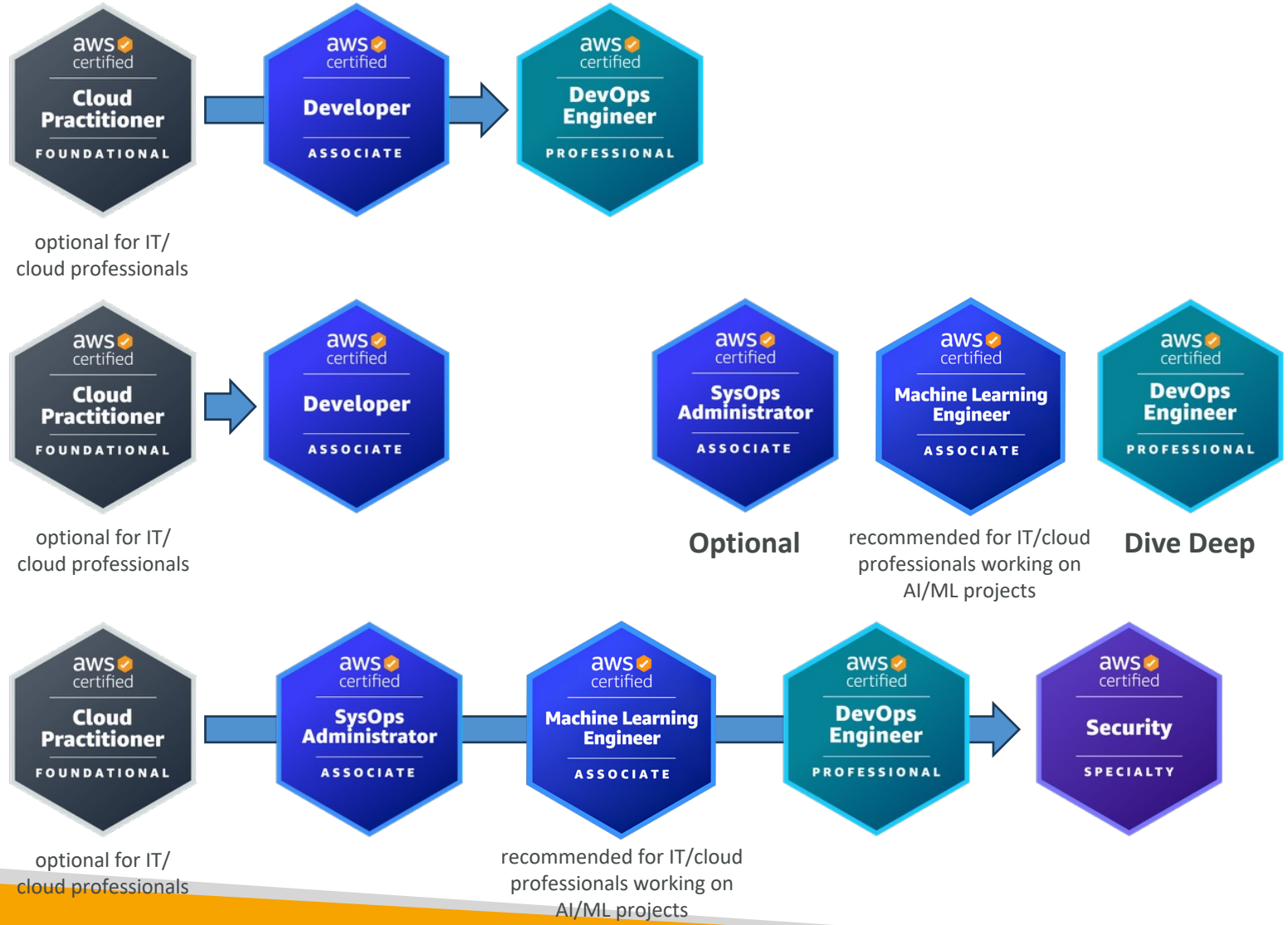
© Stephane Maarek

AWS Certification Paths – DevOps

DevOps
Test Engineer
Embed testing and quality best practices for software development from design to release, throughout the product life cycle

DevOps
Cloud DevOps Engineer
Design, deployment, and operations of large-scale global hybrid cloud computing environment, advocating for end-to-end automated CI/CD DevOps pipelines

DevOps
DevSecOps Engineer
Accelerate enterprise cloud adoption while enabling rapid and stable delivery of capabilities using CI/CD principles, methodologies, and technologies



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Security

Security

Cloud Security Engineer

Design computer security architecture and develop detailed cyber security designs. Develop, execute, and track performance of security measures to protect information



optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

Security

Cloud Security Architect

Design and implement enterprise cloud solutions applying governance to identify, communicate, and minimize business and technical risks



optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

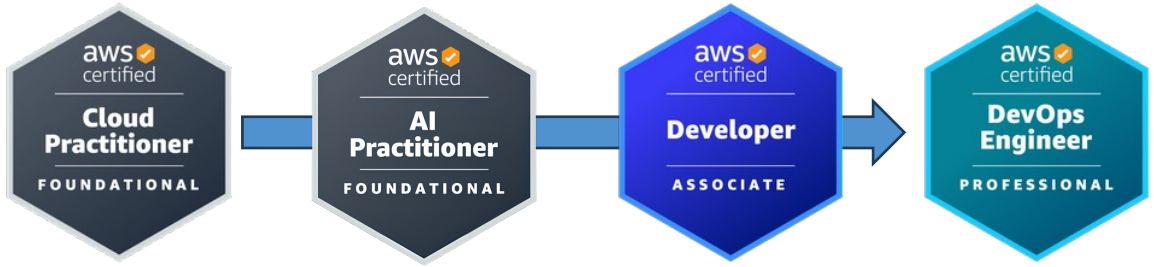
© Stephane Maarek

AWS Certification Paths – Development & Networking

Development

Software Development Engineer

Develop, construct, and maintain software across platforms and devices



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Networking

Network Engineer

Design and implement computer and information networks, such as local area networks (LAN), wide area networks (WAN), intranets, extranets, etc.



optional for IT/ cloud professionals

Dive Deep

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Certification Paths – Data Analytics & AI/ML

Data Analytics

Cloud Data Engineer

Automate collection and processing of structured/semi-structured data and monitor data pipeline performance



optional for IT/ cloud professionals

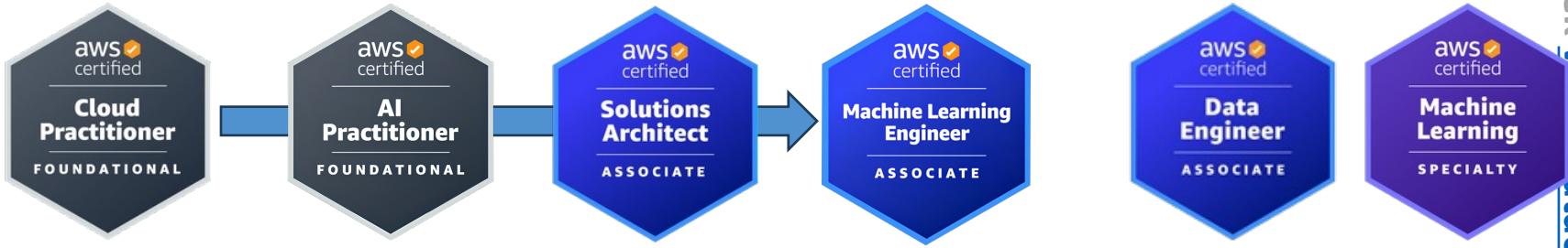
recommended for IT/cloud professionals working on AI/ML projects

Dive Deep

AI/ML

Machine Learning Engineer

Research, build, and design artificial intelligence (AI) systems to automate predictive models, and design machine learning systems, models, and schemes



optional for IT/ cloud professionals

optional for AI/ML professionals

Dive Deep

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

AWS Certification Paths – AI/ML

AI/ML

Prompt Engineer

Design, test, and refine text prompts to optimize the performance of AI language models

AI/ML

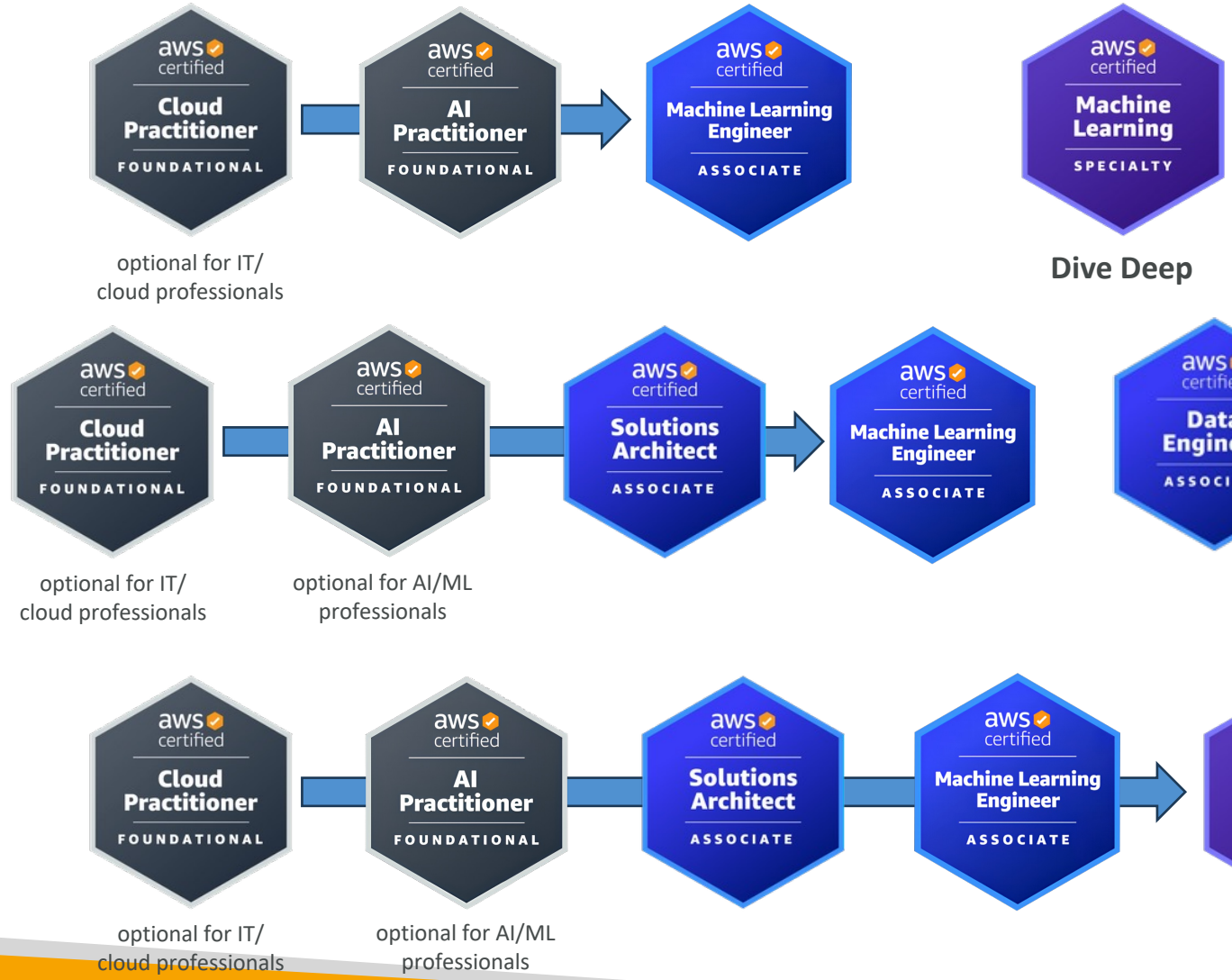
Machine Learning Ops Engineer

Build and maintain AI and ML platforms and infrastructure. Design, implement, and operationally support AI/ML model activity and deployment infrastructure

AI/ML

Data Scientist

Develop and maintain AI/ML models to solve business problems. Train and fine tune models and evaluate their performance



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com